Department of Electrical & Electronic Engineering

Rajshahi University of Engineering & Technology

LAB Report

Course Code	:	**EEE 3100**
Course Title	:	**Electronics Shop Practice**
Experiment No.	:	10
Experiment Name	:	Motor Control Using Temperature Sensor

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Date of Submission : 22 September, 2025

Experiment No.: 10

Experiment Name: Motor control using temperature sensor

Objectives:

1. To learn the fundamentals of controlling a DC motor through a temperature sensor.
2. To design and build a temperature-regulated fan system using Arduino.
3. To enable automatic adjustment of fan speed based on changes in ambient temperature.
4. To acquire hands-on experience in connecting and programming sensors and actuators with Arduino for automation purposes.
5. To enhance skills in presenting real-time data and system information on an LCD display.

Theory:

This experiment illustrates a closed-loop, temperature-controlled fan system using an Arduino Uno. An LM35 temperature sensor converts ambient temperature into an analog voltage, which the Arduino's ADC digitizes and processes into a Celsius reading. Based on this data, the Arduino regulates the speed of a DC motor proportionally using PWM signals, with an L298N motor driver providing the necessary current.

At low temperatures, the fan remains off, gradually increasing its speed as the temperature rises, mimicking the behavior of real-world cooling fans in electronics and smart systems. The project demonstrates essential concepts such as sensor interfacing, ADC conversion, data processing, PWM motor control, and feedback regulation. It effectively showcases how sensing, processing, and actuation work together to achieve efficient automation, laying a foundation for energy-efficient systems and IoT-based applications.

Required Apparatus:

Table 10.1: Required Apparatus

Sl. No.	Components	Specification	Quantity
1	Arduino Uno	ATmega328P Microcontroller	1
2	Temperature Sensor	DHT11 or LM35	1
3	DC Motor (Fan)	5V/12V	1
4	Motor Driver IC	L293D / L298N	1
5	LCD Display	16×2	1
5	Resistors, Wires	–	As per requirement
6	Breadboard	–	1
7	Power Supply	5V/12V DC	1

Circuit Diagram:

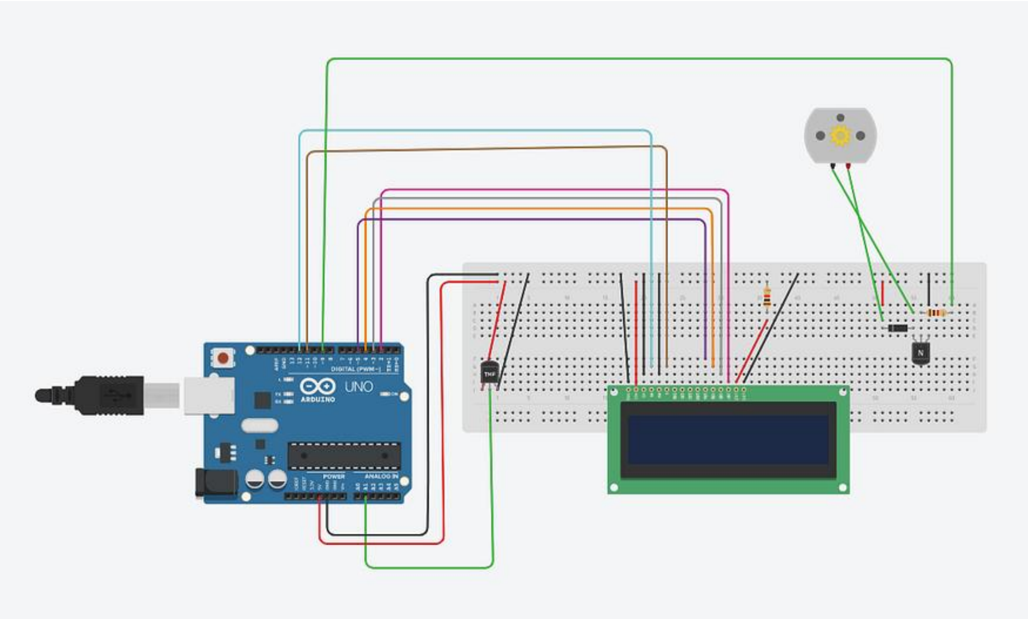

Fig.10.1: Circuit Diagram of Temperature Controlled Fan System

Experimental Setup:

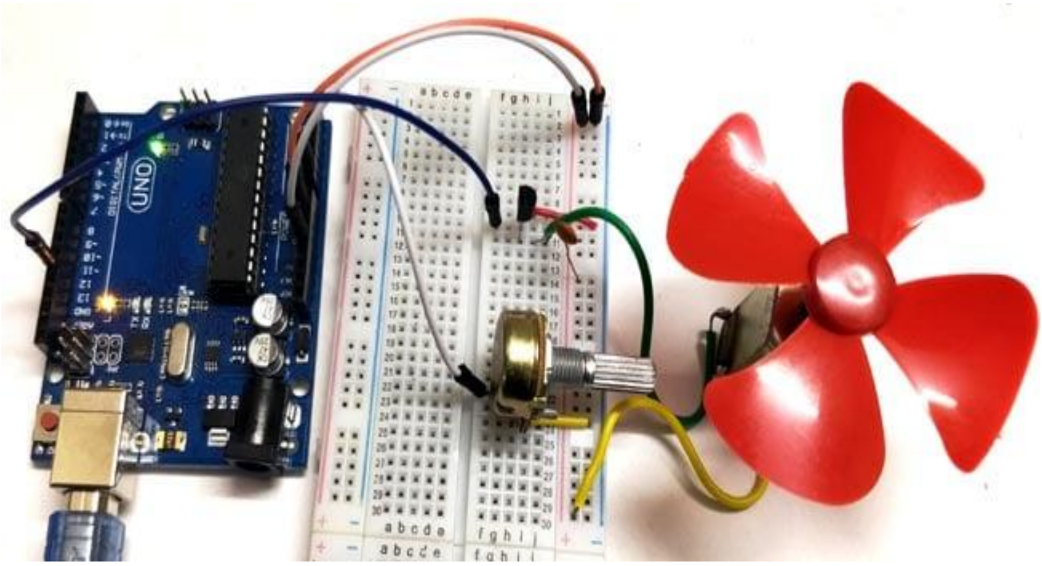

Fig.10.2: Experimental Setup of the Experiment

Code:

```
intsensorPin = A0;   // LM35 connected to analog pin A0
intmotorPin = 9;     // PWM pin for motor

voidsetup(){
  Serial.begin(9600);
  pinMode(motorPin, OUTPUT);
}

voidloop(){
  intsensorValue = analogRead(sensorPin); // Read LM35 output
  float voltage = (sensorValue * 5.0) / 1023.0; // Convert to voltage (0–5V)
  float temp = voltage * 100.0; // LM35 gives 10mV/°C → 100 * V = °C

  Serial.print("Temperature: ");
  Serial.print(temp);
  Serial.println(" °C");

  // Fan control logic
  if(temp <25){
    analogWrite(motorPin, 0);      // Fan OFF
  }elseif(temp >= 25&& temp <30){
    analogWrite(motorPin, 100);    // Low Speed
  }elseif(temp >= 30&& temp <35){
    analogWrite(motorPin, 180);    // Medium Speed
  }else{
    analogWrite(motorPin, 255);    // High Speed
  }

  delay(1000);
```

Experimental Procedure:

- **Connect** the temperature sensor, motor (through the driver), and LCD to the Arduino.
- **Upload** the program to read temperature and control the fan speed using PWM.
- **Observe** that the fan remains OFF below the set threshold and gradually increases speed as the temperature rises.
- **Record** temperature readings and corresponding motor speeds for performance analysis.

Results:

- Fan remains OFF below 25°C.
- Fan runs at **low speed** between 25°C–30°C.
- Fan runs at **medium speed** between 30°C–35°C.
- Fan runs at **full speed** above 35°C.

Discussion: The motor responded as expected to temperature fluctuations. At lower temperatures, it remained inactive, but as the heat rose, the fan speed progressively increased. There was a slight delay for the sensor readings to stabilize, but the Serial Monitor provided a convenient way to monitor the process. In summary, the system functioned effectively, demonstrating a direct relationship between temperature and motor speed.

Justification of CO2 and PO(j):

LM35 sensor to read temperature from an analog pin. The data was then used to control the speed of a motor via Pulse Width Modulation (PWM). The motor's speed increased in steps as the temperature rose. The Serial Monitor was used to track both the sensor's readings and the motor's behavior, making the process easy to follow. This experiment effectively demonstrated how input from a sensor can be used to control an output device, a concept applicable in real-world scenarios such as automatic cooling fans in laptops or smart heaters.

Conclusion: The experiment successfully demonstrated how to control a motor's speed using temperature as a variable. The fan remained off at low temperatures and increased its speed incrementally as the heat rose. The Serial Monitor was a useful tool for tracking sensor data and verifying the logic. Overall, this simple setup effectively illustrated the connection between a sensor and an actuator using an Arduino.

References:

[1] Arduino. (2023). *Analog Read Serial Example*. Arduino Official Website. Available at: https://www.arduino.cc

[2] Texas Instruments. (2016). *LM35 Precision Centigrade Temperature Sensors* [Datasheet]. Texas Instruments Incorporated. Available at: https://www.ti.com/lit/ds/symlink/lm35.pdf

[3] Mazidi, M. A., Naimi, S., & Naimi, S. (2018). *The Arduino Programming Handbook.* Pearson Education.

[4] Banzi, M., & Shiloh, M. (2014). *Getting Started with Arduino* (3rd ed.). Maker Media.